

PROJECTS & EXPERIENCE

StudyXR Framework User Research – IEEE Research Project (Press Release)

Lead Researcher & Developer | 2025

- Developed a novel XR-based framework, implementing Meta SDK and AWS in Unity3D.
- Conducted a user experience study, resulting strong positive ratings.
- Published findings in a peer-reviewed IEEE conference, contributing valuable insights to immersive technology research methodologies and user experience optimization.

Signed Distance Fields Volumetric Video Compression – IEEE Research Project (Press Release)

Developer | 2025

- Developed a novel compression algorithm for volumetric 4D video using Signed Distance Fields and Singular Value Decomposition, significantly surpassing state-of-the-art methods in compression ratio and visual quality retention.
- Published findings in a peer-reviewed IEEE conference, contributing valuable insights and advancements to the fields of volumetric video compression, real-time rendering, and immersive media technologies.

Volumetric Video Streaming Framework – IEEE Research Project (Press Release)

Lead Researcher & Developer | 2024

- Developed a cross-platform volumetric 4D video streaming solution using Python and Unity C#, implementing encoding and decoding algorithm, achieving up to 95% reduction in data size and processing over 500 streaming frames with minimal latency.
- Performed thorough performance evaluations, documenting findings and methodologies clearly for academic reporting.
- Published research findings in a peer-reviewed IEEE conference, demonstrating significant contributions to the field of real-time streaming and rendering technologies.

Teaching Assistant – Computer Graphics and Game Development

Ontario Tech University, Oshawa, ON | 2024

- Facilitated interactive workshops and lab sessions in Computer Graphics and Game Development, enhancing student engagement and comprehension regarding Unity development, programmable shader language, render pipeline and etc.
- Provided personalized tutoring and mentorship, significantly improving students' understanding of complex technical concepts.
- Collaborated closely with faculty to design rigorous assessments, effectively evaluating student performance and learning outcomes.

StudyVR Framework – IEEE Research Project (Press Release)

Lead Researcher & Developer | 2023

- Led the development of StudyVR, a Unity-based framework to streamline user studies in VR.
- Reduced study setup time by 50%.
- Designed a versatile framework suitable for various research methodologies.

Speed War – GDC and GDW Project

Lead Programmer & Project Manager | 2022

- Led a team of seven in developing a multiplayer physics-based 3D racing game using Unity C#, featuring TCP/UDP networking protocols for seamless multiplayer gameplay.
- Managed end-to-end project lifecycle, including planning, task delegation, debugging, and quality assurance, successfully delivering game features.
- Implemented advanced gameplay mechanics, steering wheel integration, and immersive haptic feedback.
- Designed intuitive GUI using Figma and Photoshop, engaging graphical elements, creating a visually appealing and user-friendly gaming environment.

Flappy Cheese – AI-Enhanced Entertainment Game

Solo Developer | 2023

- Independently developed an innovative twist on classic flappy-style gameplay using Unity C#.
- Designed and implemented reinforcement learning algorithms to train two adaptive AI agents.
- Managed end-to-end game development lifecycle, including game design, AI integration, user interface (UI), and gameplay optimization.

EDUCATION

Master of Science – Computer Science

Ontario Tech University, Oshawa ON, 2024 – Present

- Field of study: Human Perception, Computer Vision, Avatar Presentation, Immersive Technology, Human-Computer Interaction, Advanced Computer Graphics, AI and Automation, Data Analytics

Bachelor of Information Technology

Ontario Tech University, Oshawa ON, 2019 – 2023

- Specialization: Game Development and Entrepreneurship
- Field of study: Game Development & Computer Graphics

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QUALIFICATIONS

- **Software & XR Development:** Unity, Unreal, OpenGL, Blender, XR Development, Real-Time Rendering, Volumetric Video, Computer Vision, Multiplayer Networking, Machine Learning for Games
- **Programming Languages:** C#, C++, Python, JavaScript, Java, C, Swift, HTML, CSS, SQL, R, SAS
- **Front-End & UI Development:** Web Development, Three.js, Figma, UI/UX Design, Photoshop, Lightroom
- **AI & Machine Learning:** Reinforcement Learning, AI Agent Training, Python-based AI frameworks (TensorFlow, PyTorch), Prompt Engineering, Real-Time Agent in Games
- **Project Management & Collaboration:** Team Leadership, Git, Project Planning, System Design, Problem-solving, Teaching and Mentoring
- **Software & Tools:** Unity, Unreal, Git, Docker, Linux CLI, AWS, GCP, R, SAS, Blender, Substance 3D, Figma, Framer, Photoshop, Jira, Trello, Teams, Office, etc.
- **Languages:** English (Fluent), Mandarin (Native)
- **Certificates:**
 - o **Programmer Certificate in Training Certificate of National Information Technology Education Project**, Peking University Beida Jade Bird Group, Shanghai, China.